Class 9 Mathematics — Exercise 6.2

Formulae used in this exercise

$$l = r\theta \qquad A = \frac{1}{2}r^2\theta \qquad A = \frac{1}{2}rl$$

where l = length of arc, A = area of sector, r = radius and θ = central angle **in radians**.

$$1^\circ = \frac{\pi}{180}\text{ rad}, \qquad \pi \approx 3.1416$$

Question 1

Find length of arc and area of sector for given radius and central angle.

(i) $r = 5$ **cm**, $\theta = \frac{\pi}{3}$ **rad**

Length of arc:

$$l = r\theta = 5 \times \frac{\pi}{3} = \frac{5\pi}{3}$$

$$l = \frac{5(3.1416)}{3} = 5.24 \text{ cm}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(5)^2\left(\frac{\pi}{3}\right) = \frac{25\pi}{6}$$

$$A = \frac{25(3.1416)}{6} = 13.09 \text{ cm}^2$$

Answer: $l = 5.24$ cm, $A = 13.09$ cm^2

(ii) $r = 12$ **m**, $\theta = 120^\circ$

Convert the angle into radians:

$$\theta = 120 \times \frac{\pi}{180} = \frac{2\pi}{3}\text{ rad}$$

Length of arc:

$$l = r\theta = 12 \times \frac{2\pi}{3} = 8\pi$$

$$l = 8(3.1416) = 25.13 \text{ m}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(12)^2\left(\frac{2\pi}{3}\right) = \frac{1}{2}(144)\left(\frac{2\pi}{3}\right) = 48\pi$$

$$A = 48(3.1416) = 150.80 \text{ m}^2$$

Answer: $l = 25.13$ m, $A = 150.80$ m^2

(iii) $r = 6$ **dm**, $\theta = 60^\circ\, 45'\, 30''$

First convert minutes and seconds into degrees:

$$45' = \frac{45}{60} = 0.75^\circ, \qquad 30'' = \frac{30}{3600} = 0.0083^\circ$$

$$\theta = 60 + 0.75 + 0.0083 = 60.7583^\circ$$

Now convert into radians:

$$\theta = 60.7583 \times \frac{\pi}{180} = 1.0604 \text{ rad}$$

Length of arc:

$$l = r\theta = 6 \times 1.0604 = 6.36 \text{ dm}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(6)^2(1.0604) = 18 \times 1.0604 = 19.09 \text{ dm}^2$$

Answer: $l = 6.36$ dm, $A = 19.09$ dm^2

Question 2

A central angle in a circle of radius 4 cm is 75°. Find the length of the intercepted arc and area bounded by the sector.

Given: $r = 4$ cm, $\theta = 75^\circ$

$$\theta = 75 \times \frac{\pi}{180} = \frac{5\pi}{12} \text{ rad}$$

Length of arc:

$$l = r\theta = 4 \times \frac{5\pi}{12} = \frac{5\pi}{3} = \frac{5(3.1416)}{3} = 5.24 \text{ cm}$$

Area of sector:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(4)^2\left(\frac{5\pi}{12}\right) = \frac{1}{2}(16)\left(\frac{5\pi}{12}\right) = \frac{10\pi}{3}$$

$$A = \frac{10(3.1416)}{3} = 10.47 \text{ cm}^2$$

Answer: $l = 5.24$ cm, $A = 10.47$ cm^2

Question 3

Find radian measure of the central angle of a circular sector for the following data.

(i) $l = 8$ **cm**, $r = 4$ **cm**

$$l = r\theta \Rightarrow \theta = \frac{l}{r}$$

$$\theta = \frac{8}{4} = 2 \text{ rad}$$

Answer: $\theta = 2$ rad

(ii) $l = 8.5$ **m**, $r = 2.25$ **m**

$$\theta = \frac{l}{r} = \frac{8.5}{2.25} = 3.78 \text{ rad}$$

Answer: $\theta = 3.78$ rad

(iii) $A = 45$ **cm**^{2}, $r = 10.70$ **cm**

$$A = \frac{1}{2}r^2\theta \Rightarrow \theta = \frac{2A}{r^2}$$

$$\theta = \frac{2(45)}{(10.70)^2} = \frac{90}{114.49} = 0.79 \text{ rad}$$

Answer: $\theta = 0.79$ rad

(iv) $A = 100$ **cm**^{2}, $l = 10$ **cm**

Using $A = \frac{1}{2}rl$, first find the radius:

$$100 = \frac{1}{2}r\,(10) \Rightarrow 100 = 5r \Rightarrow r = 20 \text{ cm}$$

Now

$$\theta = \frac{l}{r} = \frac{10}{20} = 0.5 \text{ rad}$$

Answer: $\theta = 0.5$ rad

Question 4

Find radius of circle for the following information.

(i) $l = 4$ **cm**, $\theta = \pi$ **rad**

$$l = r\theta \Rightarrow r = \frac{l}{\theta}$$

$$r = \frac{4}{\pi} = \frac{4}{3.1416} = 1.27 \text{ cm}$$

Answer: $r = 1.27$ cm

(ii) $l = 6$ **m**, $\theta = 15^\circ$

$$\theta = 15 \times \frac{\pi}{180} = \frac{\pi}{12} \text{ rad}$$

$$r = \frac{l}{\theta} = \frac{6}{\pi/12} = \frac{72}{\pi} = \frac{72}{3.1416} = 22.92 \text{ m}$$

Answer: $r = 22.92$ m

(iii) $A = 200$ **cm**^{2}, $\theta = \frac{\pi}{4}$ **rad**

$$A = \frac{1}{2}r^2\theta \;\Rightarrow\; r^2 = \frac{2A}{\theta}$$

$$r^2 = \frac{2(200)}{\pi/4} = \frac{1600}{\pi} = \frac{1600}{3.1416} = 509.30$$

$$r = \sqrt{509.30} = 22.57 \text{ cm}$$

Answer: $r = 22.57$ cm

(iv) $A = 100$ **dm**^{2}, $l = 10$ **dm**

$$A = \frac{1}{2}rl \;\Rightarrow\; r = \frac{2A}{l}$$

$$r = \frac{2(100)}{10} = 20 \text{ dm}$$

Answer: $r = 20$ dm

Question 5

A 30 inch pendulum swings through an angle of 30°. Find the length of the arc in inches through which the tip of the pendulum swings.

The length of the pendulum is the radius of the arc.

Given: $r = 30$ inch, $\theta = 30^\circ$

$$\theta = 30 \times \frac{\pi}{180} = \frac{\pi}{6} \text{ rad}$$

$$l = r\theta = 30 \times \frac{\pi}{6} = 5\pi = 5(3.1416) = 15.71 \text{ inch}$$

Answer: $l = 15.71$ inches

Question 6

A motorcycle is traveling on a curve along a highway. The curve is an arc of a circle with radius of 10 km. If the motorcycle's speed is 42 km/h, what is the angle in degrees through which the motorcycle will turn in 21 minutes?

Given: $r = 10$ km, $v = 42$ km/h, $t = 21$ min $= \dfrac{21}{60}$ h

Distance covered along the curve (length of arc):

$$l = v \times t = 42 \times \frac{21}{60} = 14.7 \text{ km}$$

Angle turned:

$$\theta = \frac{l}{r} = \frac{14.7}{10} = 1.47 \text{ rad}$$

Convert into degrees:

$$\theta = 1.47 \times \frac{180}{\pi} = 1.47 \times 57.296 = 84.22^\circ \approx 84^\circ$$

Answer: $\theta \approx 84^\circ$

Question 7

Find perimeter and area of a half circle (diameter AB = 12 cm) by using the formulae $l = r\theta$, $A = \dfrac{1}{2}r^2\theta$.

Given: $d = 12$ cm $\Rightarrow r = 6$ cm. For a half circle $\theta = 180^\circ = \pi$ rad.

Length of curved part (arc):

$$l = r\theta = 6\pi = 6(3.1416) = 18.85 \text{ cm}$$

Perimeter = arc + diameter:

$$P = 18.85 + 12 = 30.85 \text{ cm}$$

Area:

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}(6)^2(\pi) = 18\pi = 18(3.1416) = 56.55 \text{ cm}^2$$

Answer: Perimeter = 30.85 cm, Area = 56.55 cm^2

Question 8

Find the circular measure of the angle between the hour hand and minute hand of a clock at:

Useful facts: the minute hand moves 6° per minute; the hour hand moves 30° per hour, i.e. 0.5° per minute.

(i) 9 o'clock

Minute hand is at 12 and hour hand at 9; there are 3 hour-spaces between them.

$$\text{Angle} = 3 \times 30^\circ = 90^\circ = 90 \times \frac{\pi}{180} = \frac{\pi}{2} \text{ rad}$$

Answer: $\frac{\pi}{2}$ rad

(ii) 02:30

Position of minute hand from 12: $30 \times 6^\circ = 180^\circ$

Position of hour hand from 12: $2 \times 30^\circ + 30 \times 0.5^\circ = 60^\circ + 15^\circ = 75^\circ$

$$\text{Angle} = 180^\circ - 75^\circ = 105^\circ = 105 \times \frac{\pi}{180} = \frac{7\pi}{12} \text{ rad}$$

Answer: $\frac{7\pi}{12}$ rad

(iii) 06:45

Position of minute hand from 12: $45 \times 6^\circ = 270^\circ$

Position of hour hand from 12: $6 \times 30^\circ + 45 \times 0.5^\circ = 180^\circ + 22.5^\circ = 202.5^\circ$

$$\text{Angle} = 270^\circ - 202.5^\circ = 67.5^\circ = 67.5 \times \frac{\pi}{180} = \frac{3\pi}{8} \text{ rad}$$

Answer: $\frac{3\pi}{8}$ rad